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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Half-Yearly Examination

FORM VI

MATHEMATICS 2 UNIT

Thursday 25th February 2016

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 60 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 52 Marks

- Questions 9–12 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 87 boys

Examiner

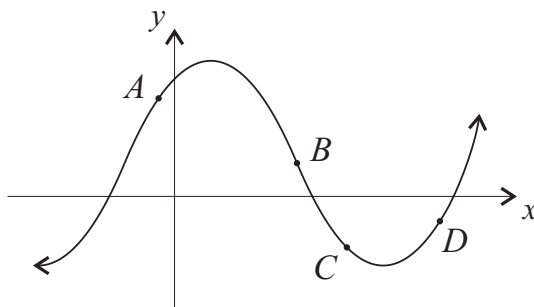
LRP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

1



The graph of $y = f(x)$ is shown above. At which point is $f'(x) > 0$ and $f''(x) > 0$?

- (A) A
- (B) B
- (C) C
- (D) D

QUESTION TWO

What is the focal length of the parabola $x^2 = 4y$?

1

- (A) $\frac{1}{4}$
- (B) 1
- (C) 4
- (D) 8

QUESTION THREE

What is the domain of the curve $y = \log_e(x - 1)$?

1

- (A) $x > -1$
- (B) $x > 0$
- (C) $x > 1$
- (D) all real x

QUESTION FOUR

Simpson's Rule is used with three function values to approximate $\int_1^2 xe^x dx$.

1

Which expression is obtained?

(A) $\frac{1}{2} (e + 3e^{1.5} + 2e^2)$

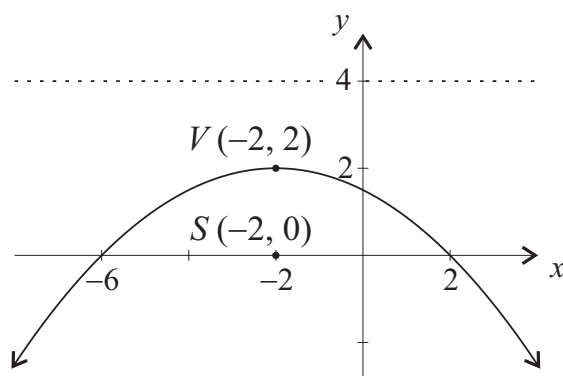
(B) $\frac{1}{4} (e + 3e^{1.5} + 2e^2)$

(C) $\frac{1}{6} (e + 6e^{1.5} + 2e^2)$

(D) $\frac{1}{12} (e + 6e^{1.5} + 2e^2)$

QUESTION FIVE

1



A parabola with focus $S(-2, 0)$ and directrix $y = 4$ is shown above. Which of the following is the equation of the parabola?

(A) $(x + 2)^2 = -8(y - 2)$

(B) $(x + 2)^2 = -8y$

(C) $(x - 2)^2 = -8(y + 2)$

(D) $(x - 2)^2 = -8y$

QUESTION SIX

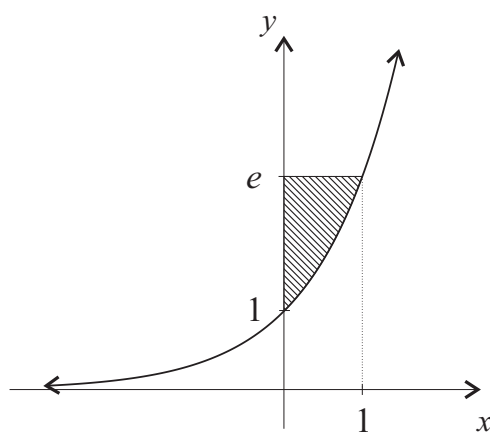
What is the value of k if $\int_1^k \frac{2}{x} dx = 4$?

1

- (A) e^2
- (B) e^3
- (C) e^4
- (D) e^5

QUESTION SEVEN

1

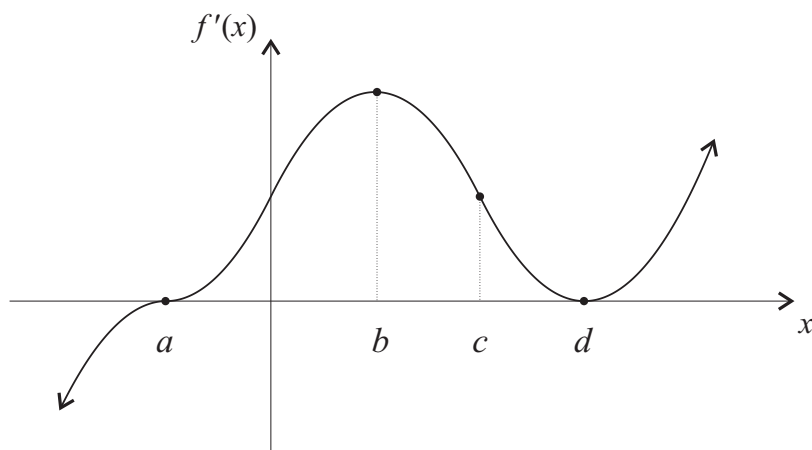


The graph of $y = e^x$ is shown above. Which of the following expressions does NOT represent the area of the shaded region?

- (A) $e - \int_0^1 e^x dx$
- (B) $\int_1^e e^x dx$
- (C) $\int_1^e \log_e y dy$
- (D) $\int_0^1 (e - e^x) dx$

QUESTION EIGHT

1



The graph of the gradient function $y = f'(x)$ is shown above. Which of the labelled x -values corresponds to a stationary point of inflexion on the original curve $y = f(x)$?

- (A) a
- (B) b
- (C) c
- (D) d

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (13 marks) Use a separate writing booklet.	Marks
(a) Calculate $\frac{3}{e^2}$ correct to 3 significant figures.	1
(b) Expand and simplify $e^x (e^x + e^{-x})$.	1
(c) Differentiate:	
(i) $\sqrt{x}$	1
(ii) e^{2x}	1
(iii) $\log_e 3x$	1
(d) Find a primitive of:	
(i) $6x^2$	1
(ii) e^{4x}	1
(iii) $\frac{3}{x}$	1
(e) Evaluate $\int_2^3 (2x - 1) dx$.	2
(f) Find the equation of the parabola with vertex $(0, 0)$ and focus $(4, 0)$.	1
(g) Consider the curve whose first derivative is given by $y' = x^2 + x - 6$. For what values of x is the curve increasing?	2

QUESTION TEN (13 marks) Use a separate writing booklet.

Marks

(a) Differentiate the following with respect to x :

(i) $y = \log_e (x^2 + 3)$

1

(ii) $y = \frac{x+1}{x}$

2

(b) Find the equation of the tangent to the curve $y = e^{3x} + 1$ at the point $(0, 2)$.

2

(c) Consider the parabola with equation $y^2 = -8(x + 2)$.

(i) Write down the coordinates of the vertex.

1

(ii) Find the coordinates of the focus.

1

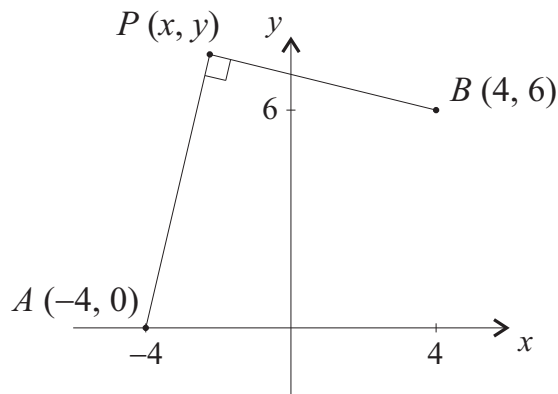
(iii) Find the equation of the directrix.

1

(iv) Sketch the parabola clearly showing the vertex, focus and directrix.

1

(d)



Consider the points $A(-4, 0)$ and $B(4, 6)$. The point $P(x, y)$ moves so that PA is always perpendicular to PB .

(i) Find expressions for the gradients of PA and PB .

1

(ii) Hence show that the equation of the locus of P is $x^2 + y^2 - 6y - 16 = 0$.

1

(iii) Express the above equation in the form $(x - h)^2 + (y - k)^2 = r^2$.

1

(iv) Hence, or otherwise, describe the locus of P geometrically.

1

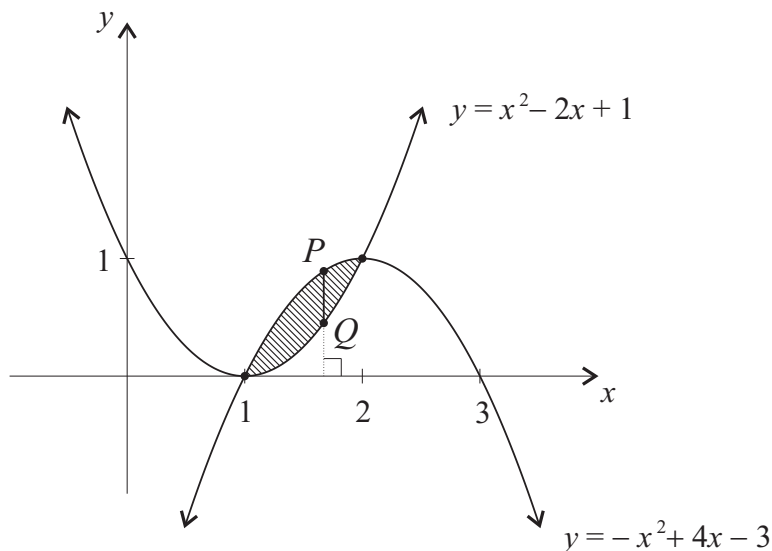
QUESTION ELEVEN (13 marks) Use a separate writing booklet.

Marks

- (a) If $f'(x) = \frac{2x-3}{x^2-3x+1}$ and $f(3) = 5$, find $f(x)$.

2

(b)



The diagram above shows the curves $y = -x^2 + 4x - 3$ and $y = x^2 - 2x + 1$.

- (i) Show that the curves intersect at the points where $x = 1$ and $x = 2$.

1

- (ii) Find the area of the shaded region enclosed between the two curves.

3

The points P and Q are located on the curves $y = -x^2 + 4x - 3$ and $y = x^2 - 2x + 1$ respectively, and share the same x -coordinate as shown.

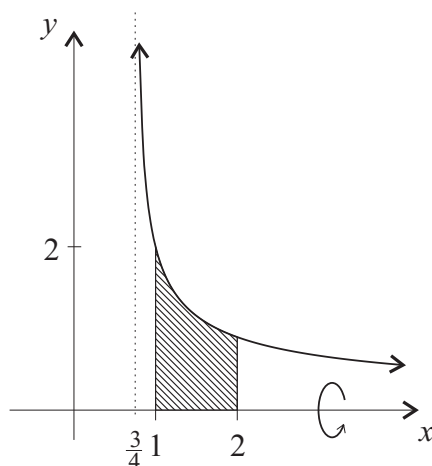
- (iii) If the length of PQ is ℓ units, show that $\ell = -2x^2 + 6x - 4$.

1

- (iv) Hence find the maximum length of PQ for $1 \leq x \leq 2$.

3

(c)



3

The region bounded by the curve $y = \frac{3}{\sqrt{4x-3}}$, the x -axis and the lines $x = 1$ and $x = 2$ is shown above. Find the volume of the solid generated when this region is rotated about the x -axis.

Exam continues next page ...

QUESTION TWELVE (13 marks) Use a separate writing booklet.

Marks

(a) Solve for x :

2

$$\log_2 (x^2 - 2x) = 3.$$

(b) (i) Show that the function $y = x \log_e x$ has derivative $\frac{dy}{dx} = \log_e x + 1$.

1

(ii) Hence find $\int \log_e x \, dx$.

2

(c) Consider the curve with equation $y = \frac{x}{e^x}$.

(i) Show that $\frac{dy}{dx} = \frac{1-x}{e^x}$ and $\frac{d^2y}{dx^2} = \frac{x-2}{e^x}$.

2

(ii) Find the coordinates of the stationary point and determine its nature.

2

(iii) Find the coordinates of the point of inflexion. You must show that it is a point of inflexion.

2

(iv) Sketch the curve, clearly showing the stationary point, the point of inflexion and any intercepts with the coordinate axes.

2

(You may use the fact that as $x \rightarrow \infty$, $\frac{x}{e^x} \rightarrow 0^+$.)

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

MATHEMATICS 2U

Half-Yearly Examination - SOLUTIONS

* Multiple Choice

Q1. D positive gradient
* concave up

Q2 B $a = 1$

Q3. C $x - 1 > 0$
 $\therefore x > 1$

Q 4. C

x	1	1.5	2
y	e	$\frac{3}{2}e^{1.5}$	$2e^2$

$$\frac{1}{e} \left(e + 4 \times \frac{3}{2} e^{1.5} + 2e^2 \right)$$

Q5. A $(x-h)^2 = -4a(y-k)$
 $(x-(-2))^2 = -4 \times 2(y-2)$

Q6. A $\int_1^k \frac{2}{x} dx = 4 \rightarrow [2 \ln x]_1^k = 4$

$$2 \ln k - 2 \ln 1 = 4$$
$$\ln k = 2$$
$$\therefore e^2 = k$$

Q7. B

Q8. D for stationary point of inflexion either

$+/- \frac{a}{+}$ or $- \frac{a}{-}$ on original curve

$\Rightarrow$ at d :  $f'(x)$ has a
 $\left. \begin{array}{l} \text{+ve value either side of } d \\ \leftarrow 0 \text{ at point } d \end{array} \right\}$

* QUESTION 9:

$$\text{a) } \frac{3}{e^2} = 0.4060058\dots \\ = 0.406 \text{ (to 3 sig. fig.)} \quad \checkmark$$

$$\text{b) } e^x(e^x + e^{-x}) = e^{2x} + e^0 \\ = e^{2x} + 1 \quad \checkmark$$

$$\text{c) i) } \frac{d}{dx}(x^{\frac{1}{2}}) = \frac{1}{2}x^{-\frac{1}{2}} \\ = \frac{1}{2\sqrt{x}} \quad \checkmark \text{ either}$$

$$\text{ii) } \frac{d}{dx}(e^{2x}) = 2e^{2x} \quad \checkmark$$

$$\text{iii) } \frac{d}{dx}(\log_e 3x) = \frac{1}{3x} \times 3 \\ = \frac{1}{x} \quad \checkmark$$

$$\text{d) i) } y' = 6x^2 \\ y = \frac{6x^3}{3} (+C) \\ = 2x^3 (+C) \quad \checkmark$$

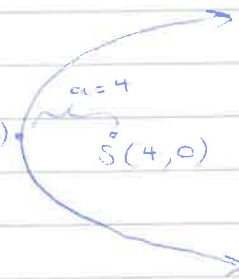
$$\text{ii) } y' = e^{4x} \\ y = \frac{e^{4x}}{4} (+C) \quad \checkmark$$

$$\text{iii) } y' = \frac{3}{x} \\ y = 3 \ln x (+C) \quad \checkmark$$

don't penalise
missing +C.

$$\begin{aligned}
 e) \int_2^3 (2x-1) dx &= \left[x^2 - x \right]_2^3 \quad \checkmark \\
 &= 3^2 - 3 - (2^2 - 2) \quad \checkmark \\
 &= 4
 \end{aligned}$$

f)



$$\begin{aligned}
 y^2 &= 4ax \\
 y^2 &= 16x \quad \checkmark
 \end{aligned}$$

g) increasing $\rightarrow y' > 0$

$$x^2 + x - 6 > 0 \quad \checkmark$$

$$(x+3)(x-2) > 0$$



$$x < -3 \text{ or } x > 2 \quad \checkmark$$

/ 13

* QUESTION 10:

$$a) i) \frac{dy}{dx} = \frac{1}{x^2+3} \times 2x \quad \checkmark$$

$$= \frac{2x}{x^2+3}$$

$$ii) y = \frac{x}{x} + \frac{1}{x}$$

$$= 1 + x^{-1} \quad \checkmark$$

$$\frac{dy}{dx} = -x^{-2} \quad \checkmark$$

$$= -\frac{1}{x^2}$$

OR //

Let

$$u = x+1$$

$$u' = 1$$

$$v = x$$

$$v' = 1 \quad \checkmark$$

$$\frac{dy}{dx} = \frac{x(1) - (x+1)(1)}{x^2}$$

$$= -\frac{1}{x^2} \quad \checkmark$$

$$b) \frac{dy}{dx} = 3e^{3x}$$

$$\text{when } x=0, \frac{dy}{dx} = 3 \quad \checkmark$$

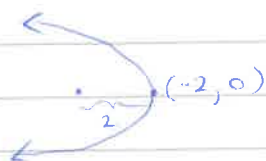
$$y-2 = 3(x-0)$$

$$\therefore y = 3x + 2 \quad \checkmark$$

c) $y^2 = -8(x+2)$

i) Vertex $(-2, 0)$ ✓

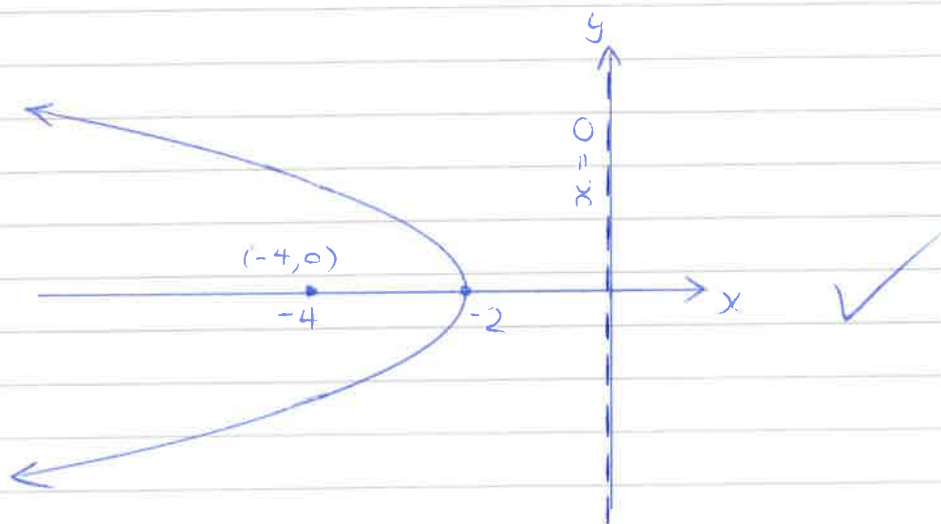
ii) $4a = 8$
 $a = 2$



Focus $(-4, 0)$ ✓

iii) Directrix $x = 0$ ✓

iv)



d) i) $m_{PA} = \frac{y}{x+4}$

$m_{PB} = \frac{y-6}{x-4}$



ii) $m_{PA} \times m_{PB} = -1$

$\frac{y}{x+4} \times \frac{y-6}{x-4} = -1$

✓ show that...

$y^2 - 6y = -x^2 + 16$

$\therefore x^2 + y^2 - 6y - 16 = 0$

iii) $x^2 + y^2 - 6y + 9 = 16 + 9$

$$x^2 + (y-3)^2 = 25$$

$$(x-0)^2 + (y-3)^2 = 5^2$$



iv) The locus of P is a circle
with centre $(0, 3)$ & radius 5 units ✓

13

* QUESTION 11:

$$a) f(x) = \log_e(x^2 - 3x + 1) + C \quad \checkmark$$

$$5 = \log_e(9 - 9 + 1) + C$$

$$5 = \log_e 1 + C$$

$$\therefore f(x) = \log_e(x^2 - 3x + 1) + 5 \quad \checkmark$$

$$b) i) x^2 - 2x + 1 = -x^2 + 4x - 3$$

$$2x^2 - 6x + 4 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0 \quad \checkmark$$

$$\therefore x = 1 \text{ or } x = 2.$$

Alternatively show the substitutions of both x values into both equations
→ four substitutions

$$ii) A = \int_1^2 (-x^2 + 4x - 3 - (x^2 - 2x + 1)) dx \quad \checkmark$$

$$= \int_1^2 (-2x^2 + 6x - 4) dx$$

$$= -2 \int_1^2 (x^2 - 3x + 2) dx$$

$$= -2 \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_1^2 \quad \checkmark$$

$$= -2 \left(\frac{8}{3} - 6 + 4 - \left(\frac{1}{3} - \frac{3}{2} + 2 \right) \right)$$

$$= \frac{1}{3} u^2 \quad \checkmark$$

$$\text{iii) } l = y_p - y_a$$

$$= -x^2 + 4x - 3 - (x^2 - 2x + 1)$$



show that...

$$= -2x^2 + 6x - 4$$

$$\text{iv) } \frac{dl}{dx} = -4x + 6$$

$$= 0 \text{ when } x = \frac{3}{2}$$



$$\frac{d^2l}{dx^2} = -4$$

$$< 0$$



$\therefore$ max occurs when $x = \frac{3}{2}$

$$\therefore l_{\max} = -2\left(\frac{3}{2}\right)^2 + 6\left(\frac{3}{2}\right) - 4$$

$$= \frac{1}{2} \text{ units}$$



Alternatively, for middle mark:

x	1	$\frac{3}{2}$	2
$\frac{dl}{dx}$	2	0	-2



$\therefore$ max occurs when $x = \frac{3}{2}$

$$\text{c) } V = \pi \int_1^2 \frac{9}{4x-3} dx$$



$$= \frac{9\pi}{4} \int_1^2 \frac{4}{4x-3} dx$$

$$= \frac{9\pi}{4} \left[\log_e(4x-3) \right]_1^2$$



$$= \frac{9\pi}{4} (\log_e 5 - \log_e 1)$$

$$= \frac{9\pi}{4} \log_e 5 \text{ u}^3$$



13

* QUESTION 12:

a) $2^3 = x^2 - 2x$ ✓

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$\therefore x = -2 \text{ or } x = 4 \quad \checkmark$$

Check $x^2 - 2x > 0$:

$$\text{When } x = -2: (-2)^2 - 2(-2) = 8 > 0$$

$$x = 4: (4)^2 - 2(4) = 8 > 0$$

Validity check not required for full marks

$\therefore$ both valid

b) i) $y = x \log_e x$

$$\text{Let } u = x \\ u' = 1$$

$$v = \log_e x \\ v' = \frac{1}{x}$$

$$y' = \log_e x \times 1 + x \times \frac{1}{x} \quad \checkmark$$

Show that...

$$= \log_e x + 1$$

ii) $\frac{d}{dx} (x \log_e x) = \log_e x + 1$

$$\therefore \int (\log_e x + 1) dx = x \log_e x + C_1 \quad \checkmark$$

$$\int \log_e x dx + \int 1 dx = x \log_e x + C_1$$

$$\therefore \int \log_e x dx = x \log_e x - \int 1 dx + C_1$$

$$= x \log_e x - x + C_2 \quad \checkmark$$

$$c) \quad y = \frac{x}{e^x}$$

$$i) \quad \text{Let } \begin{array}{ll} u = x & v = e^x \\ u' = 1 & v' = e^x \end{array}$$

$$\frac{dy}{dx} = \frac{e^x \times 1 - x \times e^x}{(e^x)^2}$$

$$= \frac{e^x(1-x)}{(e^x)^2}$$

$$= \frac{1-x}{e^x}$$

$$\text{Let } \begin{array}{ll} u = 1-x & v = e^x \\ u' = -1 & v' = e^x \end{array}$$

$$\frac{d^2y}{dx^2} = \frac{e^x \times -1 - (1-x)e^x}{(e^x)^2}$$

$$= \frac{e^x(-1-1+x)}{(e^x)^2}$$

$$= \frac{x-2}{e^x}$$

show
that...

$$ii) \quad \frac{dy}{dx} = 0 \quad \text{when } x = 1 \quad \checkmark \quad y = \frac{1}{e}$$

$$\text{when } x = 1: \quad \frac{d^2y}{dx^2} = \frac{1-2}{e^1}$$

$$= -\frac{1}{e}$$

$< 0 \quad \therefore \curvearrowright \quad \therefore \text{max t.p. occurs}$
at $(1, \frac{1}{e}) \quad \checkmark$

iii) $\frac{d^2y}{dx^2} = 0$ when $x = 2$ ✓ $y = \frac{2}{e^2}$

x	1	2	3
$\frac{d^2y}{dx^2}$	$-\frac{1}{e}$	0	$\frac{1}{e^3}$



✓
change in concavity
* point of inflexion
at $(2, \frac{2}{e^2})$

iv) when $x = 0, y = 0$

max t.p. $(1, \frac{1}{e}) \approx 0.36...$

pt of infl. $(2, \frac{2}{e^2}) \approx 0.27...$

as $x \rightarrow \infty, y \rightarrow 0^+$

